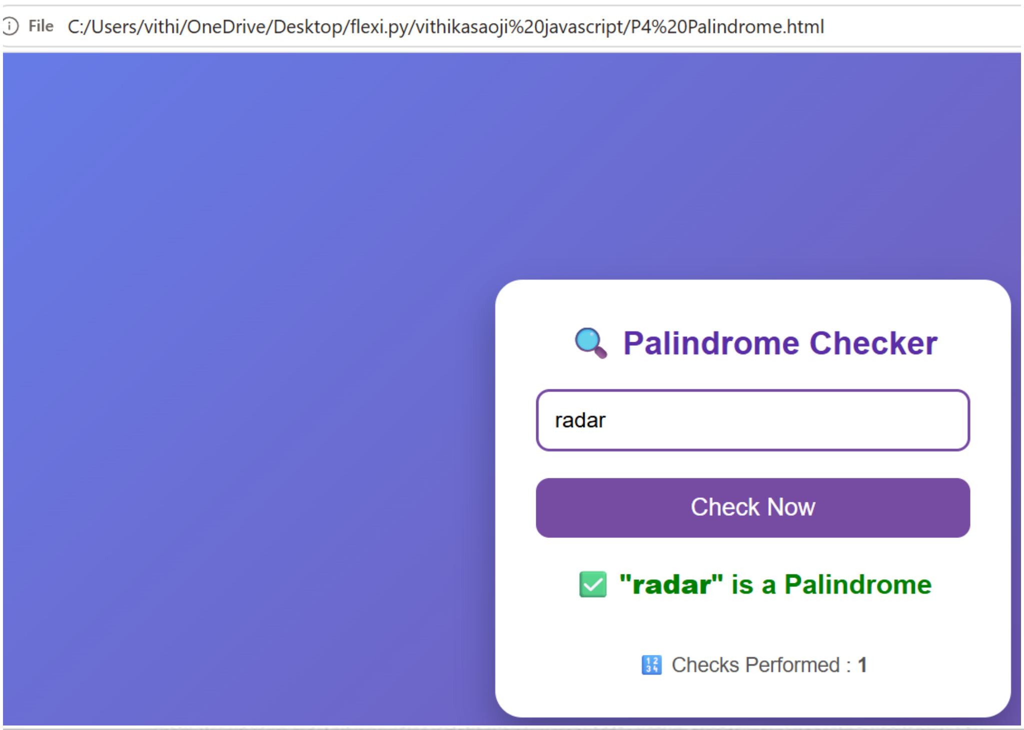
Experiment No.: 4

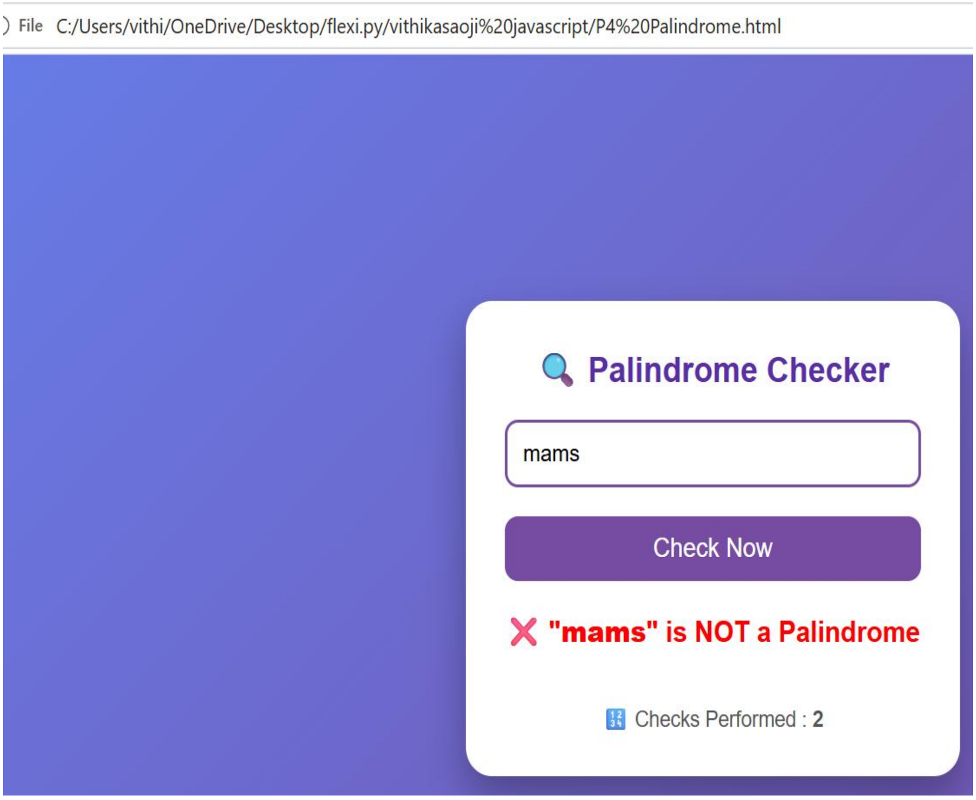
1. Title: Use function types, scope, and closures; apply try catch; build a palindrome checker.

2. Software/Tools Required: Visual Studio Code (VS Code), web browser, HTML, JavaScript.

3. Theory: This project is developed using HTML and JavaScript to create a Palindrome Checker. It demonstrates different JavaScript function types, including function declaration, function expression, and arrow functions. The program uses scope by defining global and local variables appropriately, while a closure is implemented to maintain and update the number of palindrome checks performed. Exception handling is achieved using the try...catch block, which prevents the program from crashing and displays meaningful error messages when the user enters invalid or empty input. The application accepts a word or phrase, removes unnecessary characters, checks whether it reads the same forwards and backwards, and displays the result through an interactive user interface.

4. Output :





5. Result/Conclusion: The palindrome checker was successfully developed using HTML and JavaScript. It correctly checks whether the entered text is a palindrome, handles invalid input using try-catch, demonstrates function types, scope, and closures, and displays accurate results through a simple and interactive interface. This project enhanced the understanding of core JavaScript concepts while creating a reliable, user-friendly, and error-free web application.

SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' Grade | Awarded Category – I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Case Study No.: 04

1. **Case Study Title:** ATM Card PIN Verification Using JavaScript Functions.

2. Case Study Program Code:

Release Notes: 1.131.0 | P4 Palindrome.html | P4 casestudy.html

P4 casestudy.html > html

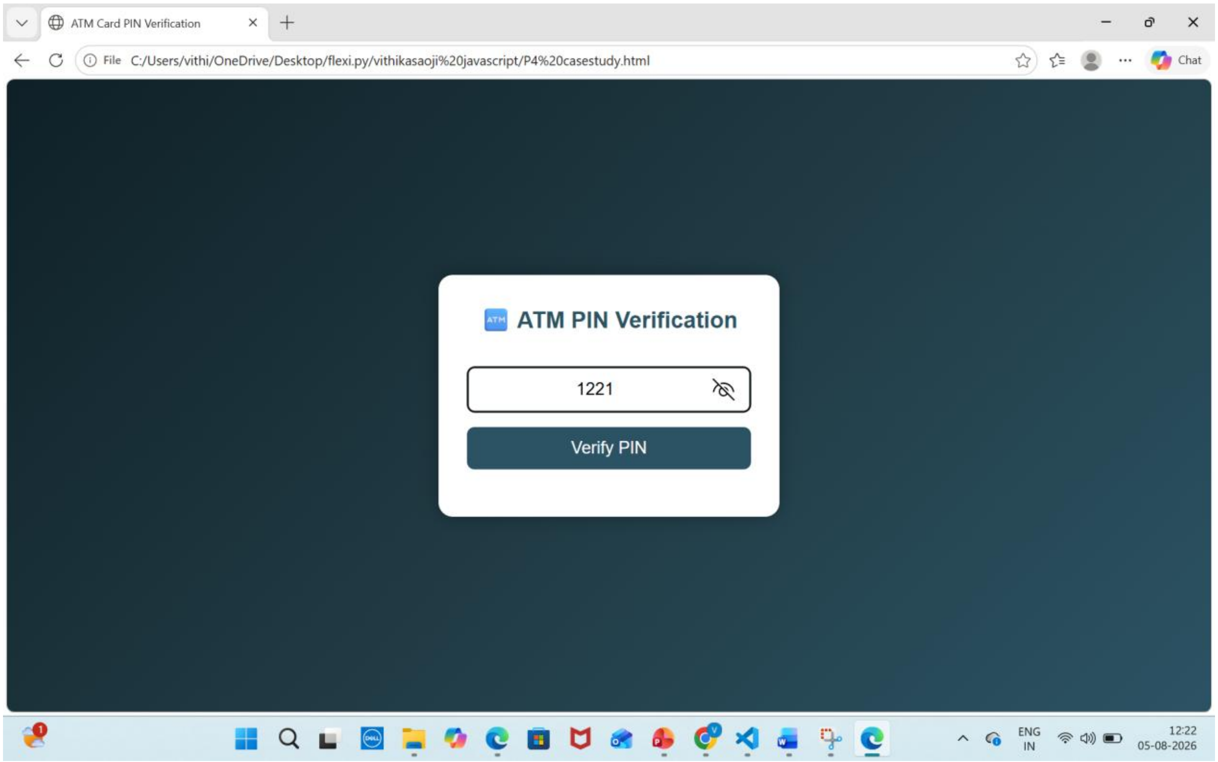
```html
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>ATM Card PIN Verification</title>

<style>
*{
    margin:0;
    padding:0;
    box-sizing:border-box;
    font-family:Arial, sans-serif;
}

body{
    height:100vh;
    display:flex;
    justify-content:center;
    align-items:center;
    background:linear-gradient(135deg,#0f2027,#203a43,#2c5364);
}

.container{
    width:360px;
    background:#fff;
    padding:30px;
    border-radius:15px;
    text-align:center;
    box-shadow:0 0 20px rgba(0,0,0,0.3);
}

h2{
    color:#2c5364;
    margin-bottom:20px;
}
```

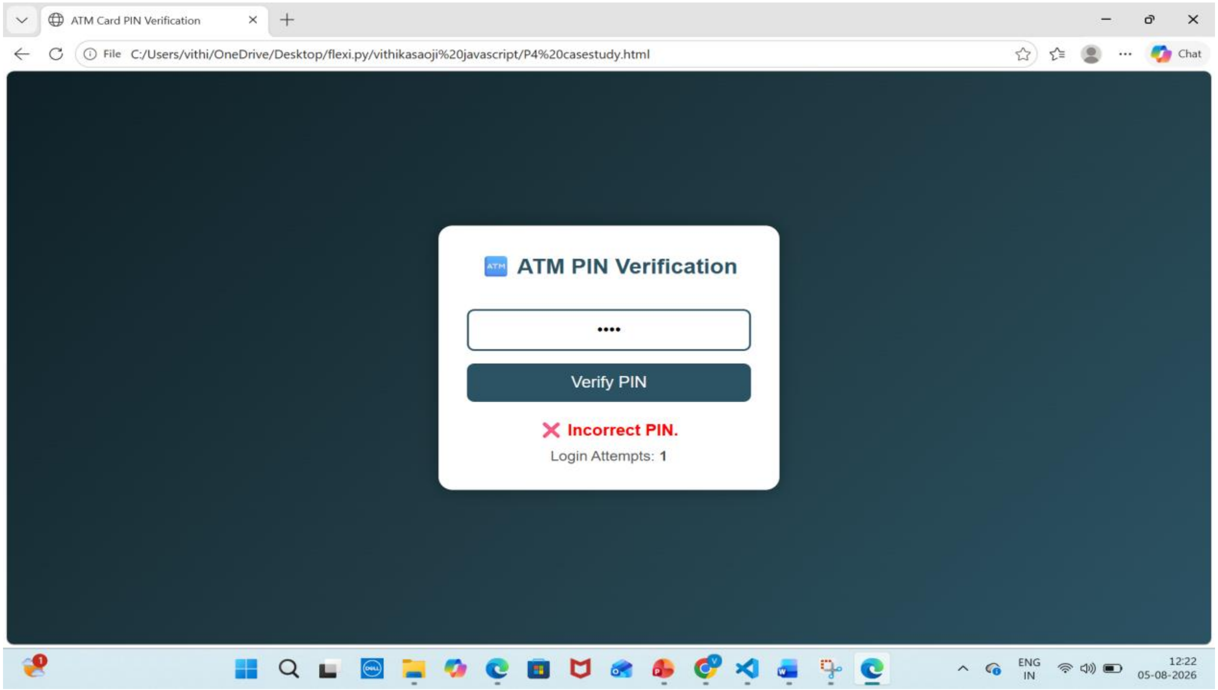
```html
input{
    width:100%;
    padding:12px;
    margin:15px 0;
    border:2px solid #2c5364;
    border-radius:8px;
    font-size:18px;
    text-align:center;
}

button{
    width:100%;
    padding:12px;
    background:#2c5364;
    color:white;
    border:none;
    border-radius:8px;
    font-size:18px;
    cursor:pointer;
}

button:hover{
    background:#1b3d4f;
}

#result{
    margin-top:20px;
    font-size:18px;
    font-weight:bold;
}

#attempts{
    margin-top:10px;
    color:#555;
}
</style>

```

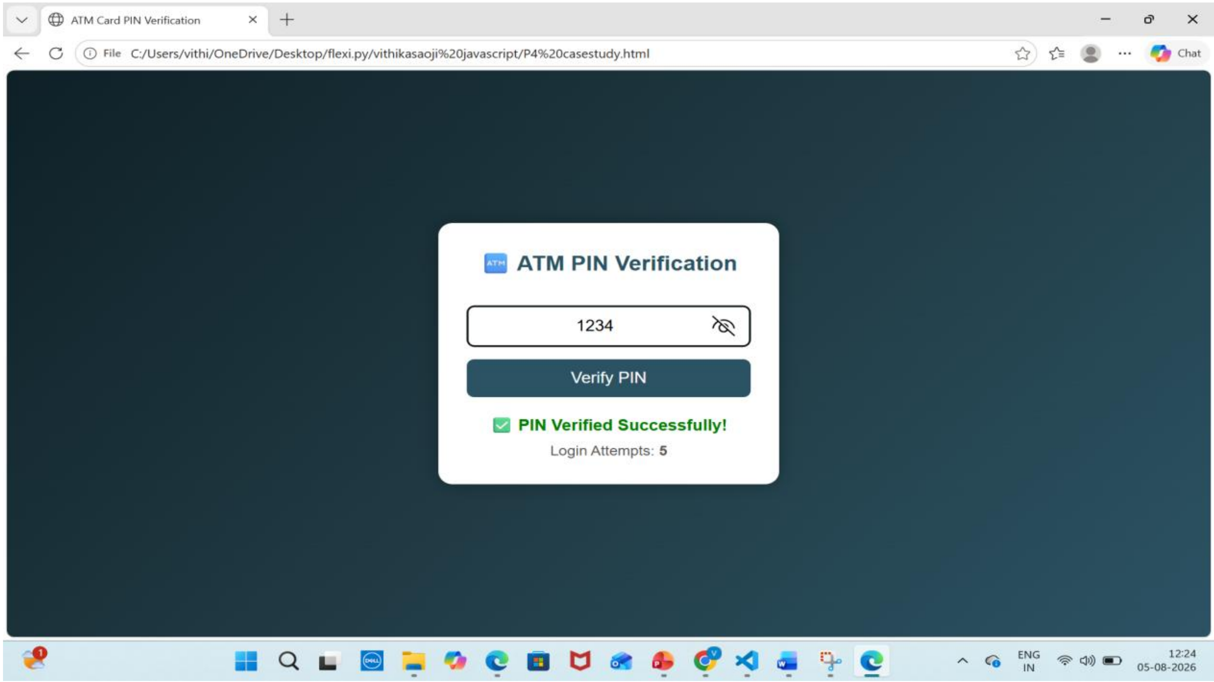
```html
</head>
<body>

<div class="container">

<h2>🏧 ATM PIN Verification</h2>

<input type="password" id="pin" maxlength="4" placeholder="Enter 4-Digit PIN">

<button onclick="verifyPIN()">Verify PIN</button>

<div id="result"></div>
<div id="attempts"></div>

</div>

<script>

// Global Scope
const correctPIN = "1234";

// Function Declaration
function getPIN(){
    return document.getElementById("pin").value;
}

// Function Expression
const checkPIN = function(pin){
    return pin === correctPIN;
};

// Closure
function loginCounter(){
    let attempts = 0;

    return function(){
        attempts++;
        return attempts;
    }
}

const countAttempts = loginCounter();

// Arrow Function
const displayMessage = (message,color)=>{
    document.getElementById("result").innerHTML = message;
    document.getElementById("result").style.color = color;
};

// Main Function
function verifyPIN(){

    try{

        let pin = getPIN();    // Local Scope

        if(pin===""){
            throw "Please enter your PIN.";
        }

        if(pin.length!=4 || isNaN(pin)){
            throw "PIN must contain exactly 4 digits.";
        }

        if(checkPIN(pin)){
            displayMessage("✅ PIN Verified Successfully!","green");
        }
        else{
            displayMessage("❌ Incorrect PIN.","red");
        }

```

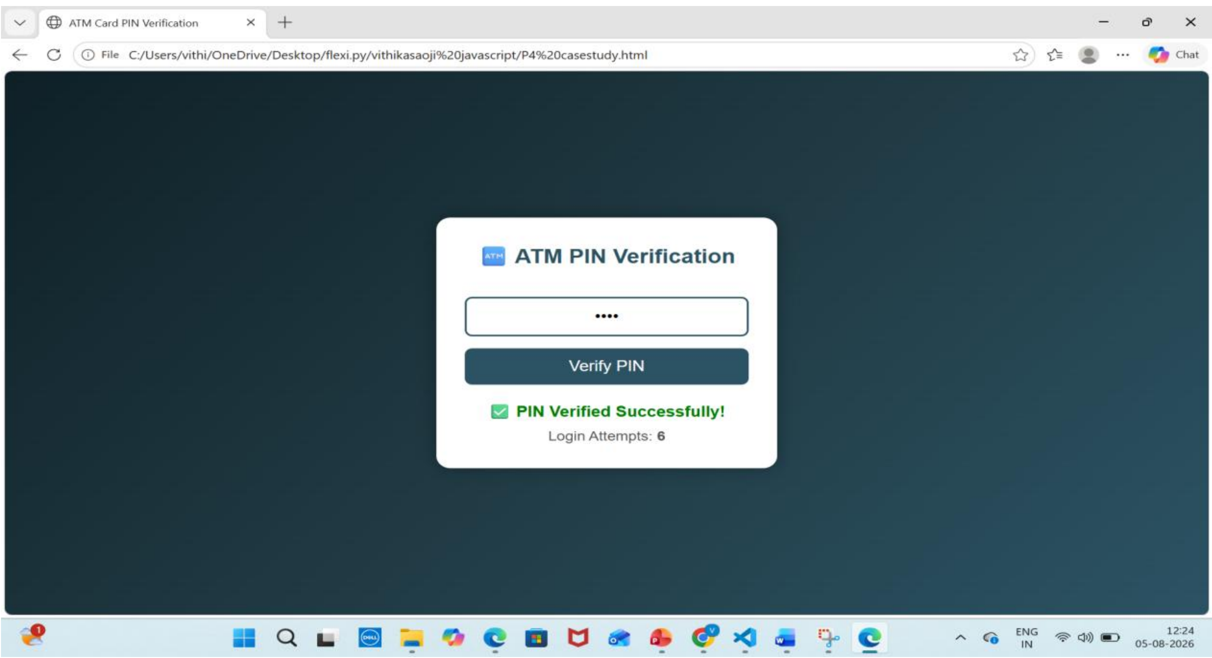
```
            }
            else{
                displayMessage("❌ Incorrect PIN.","red");
            }

            document.getElementById("attempts").innerHTML =
            "Login Attempts: <b>"+countAttempts()+"</b>";

        }

        catch(error){

            displayMessage("⚠️ "+error,"orange");

        }

        finally{

            console.log("PIN Verification Completed.");

        }

    }

</script>

</body>
</html>
```

3. Output:




ATM Card PIN Verification
File C:/Users/vithi/OneDrive/Desktop/flexi.py/vithikasaoji%20javascript/P4%20casestudy.html
Chat
ATM PIN Verification
Verify PIN
Incorrect PIN.
Login Attempts: 1
ENG
IN
12:22
05-08-2026


ATM Card PIN Verification
File C:/Users/vithi/OneDrive/Desktop/flexi.py/vithikasaoji%20javascript/P4%20casestudy.html
Chat
ATM PIN Verification
1234
Verify PIN
PIN Verified Successfully!
Login Attempts: 5
ENG
IN
12:24
05-08-2026


ATM Card PIN Verification
File C:/Users/vithi/OneDrive/Desktop/flexi.py/vithikasaoji%20javascript/P4%20casestudy.html
Chat
ATM PIN Verification
Verify PIN
PIN Verified Successfully!
Login Attempts: 6
ENG
IN
12:24
05-08-2026

4. **Result & Conclusion:** The ATM Card PIN Verification system was successfully implemented using HTML and JavaScript. It accurately verifies the entered PIN, handles invalid input using try-catch, demonstrates the use of function types, scope, and closures, and provides appropriate user feedback. This project helps in understanding core JavaScript concepts while creating a secure, reliable, and interactive PIN verification application.